

MECHANICAL GENERAL NOTES

1. INSTALL ALL WORK IN ACCORDANCE WITH THE 2018 INTERNATIONAL MECHANICAL CODE, THE 2018 INTERNATIONAL ENERGY CONSERVATION CODE, AND THE REGULATIONS OF ALL AUTHORITIES HAVING JURISDICTION.
2. TO OBTAIN ALL CONSTRUCTION INFORMATION, ALL SPECIFICATION AND DRAWINGS (E.G. ARCHITECTURAL, STRUCTURAL, CIVIL, MECHANICAL, ELECTRICAL, PLUMBING AND FIRE PROTECTION) MUST BE USED.
3. BRING TO THE ATTENTION OF THE ARCHITECT/ENGINEER ANY INFORMATIONAL CONFLICTS WITHIN THE SPECIFICATIONS AND DRAWINGS. THE CONTRACTOR(S) SHALL NOT PROCEED WITH ANY WORK UNLESS A WRITTEN UNIT OF ALL CONFLICTS ARE RESOLVED.
4. PROVIDE OFFSETS AND TRANSITIONS IN DUCTWORK AND PIPING AS REQUIRED TO AVOID INTERFERENCE AT NO ADDITIONAL COST TO THE OWNER. THOUGH SOME OFFSETS & TRANSITIONS ARE SHOWN IN PIPING & SHEET METAL TO HELP ILLUSTRATE THE PHYSICAL RELATIONSHIP BETWEEN THEM, IT IS NOT THE INTENT OF THE DRAWINGS TO SHOW ALL PIPING & SHEET METAL OFFSETS & TRANSITIONS REQUIRED. THE CONTRACTOR SHALL FULLY COORDINATE THE MECHANICAL WORK WITHIN ITS SITE WITH ALL OF THE TRADES TO PROVIDE COMPLETE & OPERABLE SYSTEMS WITHOUT INTERFERENCES.
5. INSTALL ALL EQUIPMENT LEVEL AND PLUMB. PROVIDE BLOCKING AND HARDWARE AS REQUIRED.
6. VERIFY ALL EQUIPMENT CONNECTIONS WITH MANUFACTURER'S CERTIFIED DRAWINGS. FIELD VERIFY ALL DIMENSIONS BEFORE FABRICATION OF WORK. PROVIDE TRANSITIONS AND MAKE FINAL CONNECTIONS TO EQUIPMENT. UNLESS OTHERWISE INDICATED.
7. INSTALL ALL EQUIPMENT REQUIRED PERIODIC SERVICE NOT LESS THAN 10 FEET FROM ROOF EDGE OR OPENING, UNLESS PROTECTIVE RAILING OR PARAPET OF 42 INCHES AFF. HAS BEEN PROVIDED. VERIFY WITH OTHER TRADES TO CONFIRM IF PROTECTIVE RAILING OR PARAPET OF 42 INCHES AFF IS PROVIDED.
8. PROVIDE LOADS AT 20 PSF AND REINFORCE BEAM LOCATIONS, WHERE CONCENTRATED LOADS DUE TO EQUIPMENT, RISERS, OR SPECIAL DUCT CONSTRUCTION OCCUR AT OTHER LOCATIONS. PROVIDE ADDITIONAL SUPPORT FRAMING AND/OR ANGLE BRACING.
9. DUCT CONSTRUCTION SHALL COMPLY WITH THE LATEST VERSION OF SMACNA CONSTRUCTION STANDARDS FOR THE SPECIFIC PRESSURE CLASSIFICATIONS INDICATED BELOW. ALL 90 DEGREE ELBOWS AND 90 DEGREE RADII SHALL BE USED ON ALL BEAM LOCATIONS. ALL 90 DEGREE ANGULAR ELBOWS SHALL HAVE TURNING VANS. UNLESS OTHERWISE NOTED FOR BRANCH CONNECTIONS, PROVIDE CONICAL TAPS FOR ROUND DUCTS AND 45 DEGREE TAPS FOR RECTANGULAR DUCTS.
10. FLEXIBLE DUCTWORK SHALL NOT BE USED IN RETURN OR EXHAUST DUCT SYSTEM. FLEXIBLE DUCTWORK MAY BE USED IN THE SUPPLY SYSTEM PROVIDED THE LENGTH IS LIMITED TO A TOTAL LENGTH OF FIVE (5) FEET.
11. INSTALL DUCTWORK AS CLOSE AS POSSIBLE TO UNDERSIDE OF BEAMS AND/OR JOISTS.
12. ALL DUCTWORK SIZES INDICATED ARE INSIDE CLEAR DIMENSIONS.
13. PROVIDE DUCTWORK TO TERMINAL AIR DEVICES AT SCHEDULED INLET SIZE UNLESS OTHERWISE INDICATED.
14. PROVIDE PENETRATIONS OF ALL FIRE RATED AND SMOKE RESISTIVE CONSTRUCTION TO COORDINATE WITH LIFE SAFETY PLANS FOR LOCATIONS. PENETRATION ASSEMBLY TO BE U.L. LISTED.
15. PAINT FLAT BLACK THE INSIDE OF ALL DUCTWORK VISIBLE THROUGHOUT THE SYSTEM. GRILLS AND REGISTER MOUNTING FRAME TYPES WITH THE CEILING TYPES. PROVIDE FRAME TYPE THAT MATCHES CEILING CONFIGURATION.
16. PROVIDE OPEN END DUCTWORK (JED) WITH 22 MESH (1/2" X 1/2") ALUMINUM SCREEN WITH REMOVABLE FRAMES.
17. FURNISH AND INSTALL ALL MANUALLY OPERATED VOLUME CONTROL (BALANCING) DAMPERS NECESSARY TO PERFORM THE BALANCING WORK, AT EACH BRANCH TAKEOFF, AT EACH AIR TERMINAL DEVICE AND WHERE INDICATED.
18. DUCT SYSTEMS SHALL BE BALANCED TO APOLOWS DRAWING. FANS SHALL BE FIELD TESTED TO ENSURE COMPLIANCE WITH SCHEDULED FAN PERFORMANCE AND SCHEDULED AIRFLOW AT THE DESIGN STATIC PRESSURES LISTED.
19. PROVIDE FLEXIBLE CONNECTIONS BETWEEN DUCTWORK AND ALL FANS AND ALL HANDLING EQUIPMENT.
20. VERIFY ALL DUCT CONNECTIONS, RISERS, GRILLS AND REGISTERS WITH THE REFLECTED CEILING PLANS BEFORE INSTALLATION. PROVIDE SYMMETRY WITH THE ROOM LIGHTING SYSTEMS.
21. VERIFY DIFFUSER, GRILLE AND REGISTER MOUNTING FRAME TYPES WITH THE CEILING TYPES. PROVIDE FRAME TYPE THAT MATCHES CEILING CONFIGURATION.
22. PROVIDE ACCESS DOORS AND/OR PANELS FOR DUCT MOUNTED FIRE, SMOKE, FUEL, SMOKE DAMPERS AND SMOKE DETECTORS.
23. PROVIDE ACCESS DOORS AND/OR PANELS IN HAND CEILINGS AND WALLS TO PERMIT SERVICE TO DAMPERS, VALVES, AND CONCEALED EQUIPMENT.
24. REFUSE TO ACCEPT CONTRACTOR'S GOVERNANCE INFORMATION OR SUE FOR MOUNTING HEIGHTS OF DEVICES, WHERE THIS INFORMATION IS NOT PROVIDED, MOUNT ROOM THERMOSTATS AT 48 INCHES AFF AND OCCUPANCY SENSORS AT 80 INCHES AFF UNLESS OTHERWISE INDICATED.
25. COORDINATE ALL WALL MOUNTED DEVICES SUCH AS THERMOSTATS AND CONTROL PANELS WITH ARCHITECTURAL INTERIOR ELEVATIONS.
26. COORDINATE ALL DOOR UNDER CUTS INDICATED AND AS REQUIRED WHERE RETURN/TRANSFER OPENS ARE NOT PROVIDED.
27. VERIFY ALL SNAKE RACKS SHALL BE SUBMITTED BY CONTRACTOR TO THE AUTHORITY HAVING JURISDICTION BEFORE FINAL INSPECTION SCHEDULED.
28. HVAC AUTOMATIC TEST REPORT FOR UNITS WITH DUCTWORK SMOKE DETECTORS SHALL BE SUBMITTED BY CONTRACTOR TO THE AUTHORITY HAVING JURISDICTION.
29. RETURN AIR CMFS LISTED ON DRAWINGS ARE APPROXIMATIONS BASED ON INITIAL DESIGN CALCULATIONS TO ADDRESS PRESSURE RELATIONSHIPS BETWEEN ADJACENT SPACES, FLOORS, AND THE OUTDOORS. FINAL RETURN AIR CMFS MAY VARY FROM LISTED RETURN AIR CMFS. FINAL AIR BALANCE TESTING SHALL DETERMINE FINAL RETURN AIR CMFS. SUCH PRESSURE RELATIONSHIPS ARE NOT GUARANTEED BETWEEN ADJACENT SPACES, FLOORS, AND THE OUTDOORS.
30. THERMOSTATIC CONTROLS HAVE A 5% DEADBAND, VAV UNIT SETBACK TO 55° (F) AND 85° (C); 7.04K COIL; 2.04H COIL OCCUPANT, 10-HOUR BACK UP. GRAC UNITS AND SPLIT SYSTEM SHALL NOT HAVE A SETBACK.
31. IF MANUFACTURER'S SOURCE, WHERE A SINGLE PRODUCT OR MANUFACTURER'S NAME IS NAMED, COMPARABLE PRODUCTS COMPLIANT WITH REQUIREMENTS WILL BE CONSIDERED, UNLESS ACCOMPANIED BY THE PHRASE "NO KNOWN EQUIV."

DUCT SYSTEM	LOCATION		PRESSURE CLASSIFICATION (INCH WATER GAUGE)	DUCTWORK SEAL/LEAKAGE CLASS	REMARKS
	FROM	TO			
GENERAL SUPPLY	SUPPLY RISER	DIFFUSERS, REGISTERS	3	A	
GENERAL RETURN	RETURN RISER	DIFFUSERS, REGISTERS	-2	A	
GENERAL EXHAUST	THROUGH OUT		-2	A	

REFERENCE SYMBOLS



SECTION NUMBER _____

DRAWING NUMBER WHERE DRAWN _____



_____ RISER DESIGNATION (TYPE AS INDICATED)



DETAIL NUMBER _____

DRAWING NUMBER WHERE DRAWN _____



_____ KEY NOTE



_____ DEMO KEY

SECTION NUMBER
DRAWING NUMBER WHERE DRAWN
HHWS
4
RISER DESIGNATION (TYPE AS INDICATED)
DETAIL NUMBER
DRAWING NUMBER WHERE DRAWN
MOXY
KEY NOTE
4
DEMO KEY

SHEET NUMBER

M0.1
